

1. Contextualization of the Capstone Project

1.1 What is the project?

This project was developed as part of a challenge proposed by **Escola da Nuvem**, aiming to practicalize and reinforce knowledge acquired about cloud computing—specifically how AWS services can be leveraged to solve real-world business problems.

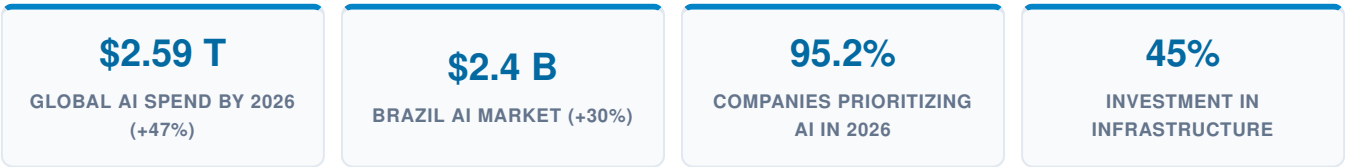
The challenge offers an opportunity to move beyond purely conceptual models and apply the competencies studied during preparation for the **AWS Certified Cloud Practitioner** certification to analyze and design a robust architectural solution.

Beyond simply selecting AWS services, the objective is to deeply understand the business problem, map operational constraints, and evaluate how cloud-native resources create tangible business value. This perspective highlights that the system involves much more than raw file storage. Documents possess a distinct lifecycle, varying access frequencies over time, strict security requirements, and long-term strategic value. Consequently, decisions regarding data organization, access control, retention, and cost optimization are foundational to the solution architecture.

1.2 Why is this topic relevant?

This theme is particularly relevant given the exponential growth of Artificial Intelligence (AI) and the critical importance of the data pipelines fueling these models. In the proposed scenario, documents uploaded by clients form the historical repository required for training future AI models. Ensuring this data remains organized, secured, highly available, and economically sustainable over time is central to the enterprise's strategic longevity.

Furthermore, the project operates within a rapidly expanding global AI ecosystem, backed by strong industry metrics:



- **Global Growth:** According to Gartner projections, global investment in artificial intelligence will reach \$2.59 trillion in 2026—a 47% increase year-over-year—with expectations to approach nearly \$3.5 trillion by 2027.
- **National Scenario (Brazil):** An IBM study revealed that 78% of Brazilian companies plan to expand AI investments, with national AI expenditures surpassing \$2.4 billion (a 30% surge compared to 2024).
- **Strategic Adoption:** Enterprise adoption has surged, with 95.2% of organizations considering AI a core strategic priority for 2026, up from just 32.8% actively investing in 2024.
- **Infrastructure Focus:** Approximately 45% of total projected AI investments in 2026 are dedicated to AI infrastructure, including cloud storage, compute capabilities, and data pipelines.

"All AI evolution relies on a fundamental requirement: well-stored, organized, protected, and accessible data. High-quality AI cannot exist without a robust historical dataset behind it."

Thus, the project addresses a pivotal challenge: *How to build an infrastructure capable of converting expanding data volumes into a strategic enterprise asset without compromising security, availability, or financial sustainability?*

2. Project Context & Business Case

Startup Jarva's is a hyper-growth enterprise offering a SaaS platform dedicated to intelligent data extraction driven by Artificial Intelligence.

As part of its core workflow, the platform receives user-submitted documents (primarily PDFs and images) that serve as raw input for AI ingestion. The platform anticipates an initial volume of approximately **50,000 new files per month (~600,000 files/year)**, with sustained compound growth.

Critical Business Mandate: *No documents may ever be deleted.* Historical records carry immense intrinsic value as they constitute the foundation for training future proprietary AI models. As the business scales, data volume expands continuously into millions of records. This repository represents not merely data storage, but an irreplaceable intellectual asset driving the company's competitive edge.

3. Problem Statement & Business Impact

The central architectural challenge stems from combining rapid operational growth, strict multi-tenant customer isolation, mandatory permanent retention, and strict storage cost controls.

3.1 Data Security & Multi-Tenant Isolation

The platform operates on a multi-tenant architecture where diverse client accounts share common application layers. This necessitates absolute, airtight logical isolation: *no client may ever view or access another client's documents.*

Users must retrieve their own files seamlessly and rapidly, without risk of misconfiguration, privilege escalation, or permission flaws exposing cross-tenant data. Tenant isolation must be natively designed into the architecture from day one.

Risk Perspective: High-profile historical data breaches (such as the widely reported 2022 Elasticsearch exposures) underscore the severe liability of flawed cloud storage permissions. Faulty configurations can transform secure repositories into public liabilities.

3.2 Document Retention & Storage Cost Scaling

Unlike conventional systems where legacy data is purged, Startup Jarva's mandates zero operational deletion. Storing 50,000+ new files monthly on high-performance primary storage indefinitely would produce exponential, unsustainable operational expenditures.

To solve this, the architecture implements automated data lifecycle management via **Amazon S3 Lifecycle Rules**:

- **First 12 Months:** Documents reside in **Amazon S3 Intelligent-Tiering**, optimizing costs across frequent and infrequent access patterns with instant availability.
- **After 12 Months:** An automated S3 Lifecycle rule transitions objects to **Amazon S3 Glacier Deep Archive**, drastically lowering long-term storage costs while preserving data integrity.

3.3 The Core Architectural Dilemma

The solution must simultaneously satisfy four key operational pillars:

1. **Rapid Access:** Instant, low-latency document retrieval during the initial 12-month active window.
2. **Permanent Retention:** Immutable long-term preservation without operational deletion.
3. **Cost Control:** Automated tier transitions to optimize storage expenditure at scale.
4. **Isolation & Security:** Strict, verified multi-tenant data privacy.

4. Project & Solution Requirements

- ✓ **Security:** Rigorous multi-tenant isolation and principle-of-least-privilege access controls.
- ✓ **Scalability:** Native elasticity supporting 50k+ files/month without manual server provisioning.

- ✓ **Durability & Retention:** Permanent data protection against accidental or malicious deletion.
- ✓ **Availability & Performance:** High-speed access during active usage windows (first 12 months).
- ✓ **Cost Efficiency:** Automated lifecycle transitions lowering storage costs as files age.
- ✓ **Operations & Observability:** Full traceability, central logging, and active budget alerting.
- ✓ **Reproducibility:** 100% Infrastructure as Code (IaC) via AWS CloudFormation templates.
- ✓ **Resilience:** Fully serverless design eliminating single points of failure (SPOFs).

5. Proposed Architectural Solution

5.1 Overview & Serverless Paradigm

The proposed solution employs a fully serverless, cloud-native architecture built on managed AWS services. It decouples authentication, authorization, business logic, object storage, metadata indexing, and system observability.

Data transfers utilize temporary **S3 Pre-Signed URLs**. Clients upload and download files directly to/from Amazon S3, bypassing API Gateway and Lambda for heavy payload transfers. This eliminates compute bottlenecks, minimizes latency, lowers bandwidth costs, and enforces security.

5.2 AWS Services Breakdown

AWS Service	Role in the Architecture
Amazon Cognito	User identity management, secure authentication, and JWT token issuance.
Amazon API Gateway	RESTful API entry point, token validation, throttling, and Lambda integration.
AWS Lambda	Serverless execution of business logic, authorization verification, and Pre-Signed URL generation.
Amazon DynamoDB	NoSQL database storing document metadata, ownership mappings, and access controls.
Amazon S3 Intelligent-Tiering	Primary storage tier dynamically optimizing cost based on access patterns over the first 12 months.
S3 Lifecycle Rules	Automated rules transitioning objects older than 12 months to cold archive tiers.
S3 Glacier Deep Archive	Ultra-low-cost storage tier for long-term historical preservation.
S3 Object Lock	WORM (Write Once, Read Many) immutability protection preventing improper modification or deletion.
AWS IAM	Fine-grained identity policies enforcing Least Privilege Access across all components.
Amazon CloudWatch	Centralized monitoring, operational logs, custom metrics, and metric alarms.
AWS CloudTrail	Comprehensive audit trail logging all API activity across the AWS account.
AWS Budgets & Cost Explorer	Active financial monitoring, cost forecasting, and automated budget threshold alerts.
AWS CloudFormation	Declarative Infrastructure as Code (IaC) templates for automated, repeatable deployment.

5.3 Customer Journey (End-to-End Workflow)

The customer journey seamlessly integrates security validation, efficient file ingestion, and automated data archiving:

1. **Authentication:** The user authenticates via Amazon Cognito, obtaining a secure JWT token.
2. **Upload Request:** The client app requests an upload link by calling API Gateway with file metadata.

3. **Validation & Metadata:** Lambda validates the request, writes ownership metadata into DynamoDB, and generates a scoped S3 Pre-Signed Upload URL.
4. **Direct S3 Upload:** The client uploads the file directly to Amazon S3 Intelligent-Tiering using the pre-signed URL.
5. **Document Access:** For retrieval, Lambda verifies user ownership via DynamoDB and returns a temporary S3 Pre-Signed Download URL.
6. **Automated Archiving:** After 12 months, S3 Lifecycle rules automatically transition the file to S3 Glacier Deep Archive, locking in minimal long-term storage costs.

5.4 Technical Architecture Highlights

- **Multi-Tenant Isolation:** DynamoDB indexes partition key by `TenantID / UserID`, ensuring queries can only return objects belonging to the authenticated principal.
- **Zero-Server Overhead:** Fully managed serverless services automatically scale from zero to peak demand without infrastructure provisioning or server maintenance.
- **High Durability & Availability:** Managed S3 and DynamoDB replicate data redundantly across multiple Availability Zones (AZs), delivering 99.999999999% (11 9's) data durability.

5.5 Governance, Security & Best Practices

- **AWS Well-Architected Alignment:** Designed according to the 6 pillars—Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.
- **AWS Cloud Adoption Framework (CAF):** Incorporates best practices for security posture, cloud governance, and operational readiness.

6. How the Project Was Developed

Development followed a structured 4-phase engineering lifecycle:

1. **Requirements Mapping & Problem Analysis:** In-depth assessment of data growth (~50k files/month), access patterns, and cost/retention constraints.
2. **Architectural Design:** Blueprinting the serverless architecture, selecting direct S3 pre-signed transfers to maximize throughput and eliminate API bottlenecks.
3. **Data Modeling & Security Policy:** Schema design in DynamoDB for metadata/ownership, coupled with tight IAM policy definitions.
4. **Infrastructure as Code (IaC):** Authoring modular AWS CloudFormation templates to provision all resources automatically in a repeatable, version-controlled manner.

7. Infrastructure as Code (IaC)

By defining 100% of the architecture in **AWS CloudFormation** templates, Startup Jarva's achieves complete environment reproducibility. Development, staging, and production environments can be spun up consistently without human error, facilitating CI/CD pipelines, strict governance, and full version control of infrastructure assets.

8. Evolution & Future Vision — Phase 02

Planned enhancements for Phase 02 include:

- **AWS KMS Integration:** Customer-Managed Keys (CMK) for enterprise-grade, envelope encryption meeting strict compliance standards.
- **Amazon EventBridge & AWS Step Functions:** Asynchronous, orchestrated pipelines for automated text extraction, OCR, and AI indexing upon upload.

- **Amazon CloudFront & S3 Transfer Acceleration:** Global CDN edge caching and accelerated upload endpoints for low-latency access by international users.
- **Automated AI Training Pipelines:** Event-driven batch extraction from Glacier archives to continuously retrain and refine custom Machine Learning models.

9. Acknowledgments

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10. References

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- Gartner. *Forecast: Enterprise Information Technology Services, Worldwide (2024–2027)*.
- IBM Institute for Business Value. *Study on Artificial Intelligence Adoption in Enterprise Systems*.
- Official Project Repository: <https://github.com/lbarretto/jarvas-aws-architecture>